

While loop



Agenda

- ① Iterative control
- ② while
- ③ Practice problem

Iterative Control (Loop)

① while

② for

X do while



Keyword

while

False



while condition :

code
code
code

True



code



Write a program to print **pws**
five times.

$i = 1$ $\leftarrow$ initialization
while $i \leq 5$: $\leftarrow$ condition

print("pws")
 $i += 1$ $\leftarrow$ flow

Write a program to print first 10 natural numbers.

1 2 3 4 5 6 7 8 9 10

```
i = 1
```

```
while i <= 10:
```

```
    print(i, end = ' ')
```

```
    i = i + 1
```

① Read/understand

② Test Case

③ Dry Run

Write a program to print first 10 natural numbers in reverse order

10 9 8 7 6 5 4 3 2 1

$i = 10$

while $i \geq 1$:

print(i , end=' ')

$i -= 1$

$i = 1$

while $i \leq 10$:

print($11 - i$)

$i += 1$

i	$11 - i$
1	10
2	9
3	8
4	7
...	...
10	1

Write a program to print first n even natural numbers.

$n=5$

2 4 6 8 10

$n=6$

2 4 6 8 10 12

$n=10$

2 4 6 8 10 12 14 16 18 20

1 2 3 4 5 6 7 8 9 10 11 12

$i=2$

$n = \text{int}(\text{input}(\text{"Enter a num"}))$

while $i \leq 2 * n$:

print(i , end='')

$i = i + 2$